

S7CAS-PLEIADES-CARTE-MERE

Pleiades

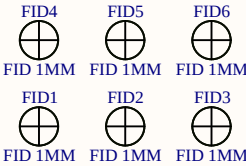
01

Revision 1.00

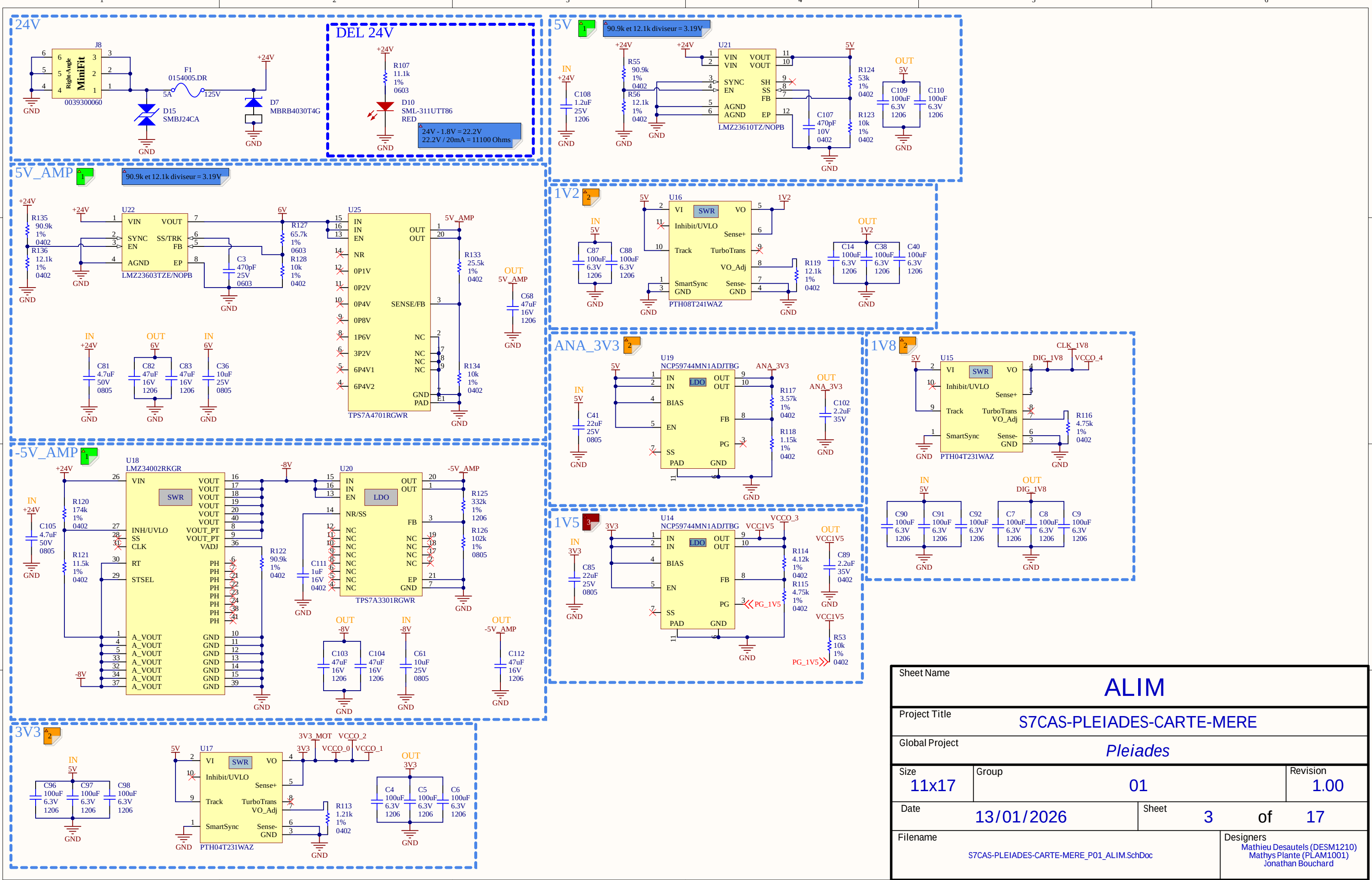
Date : 13/01/2026

Revision history	

Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512

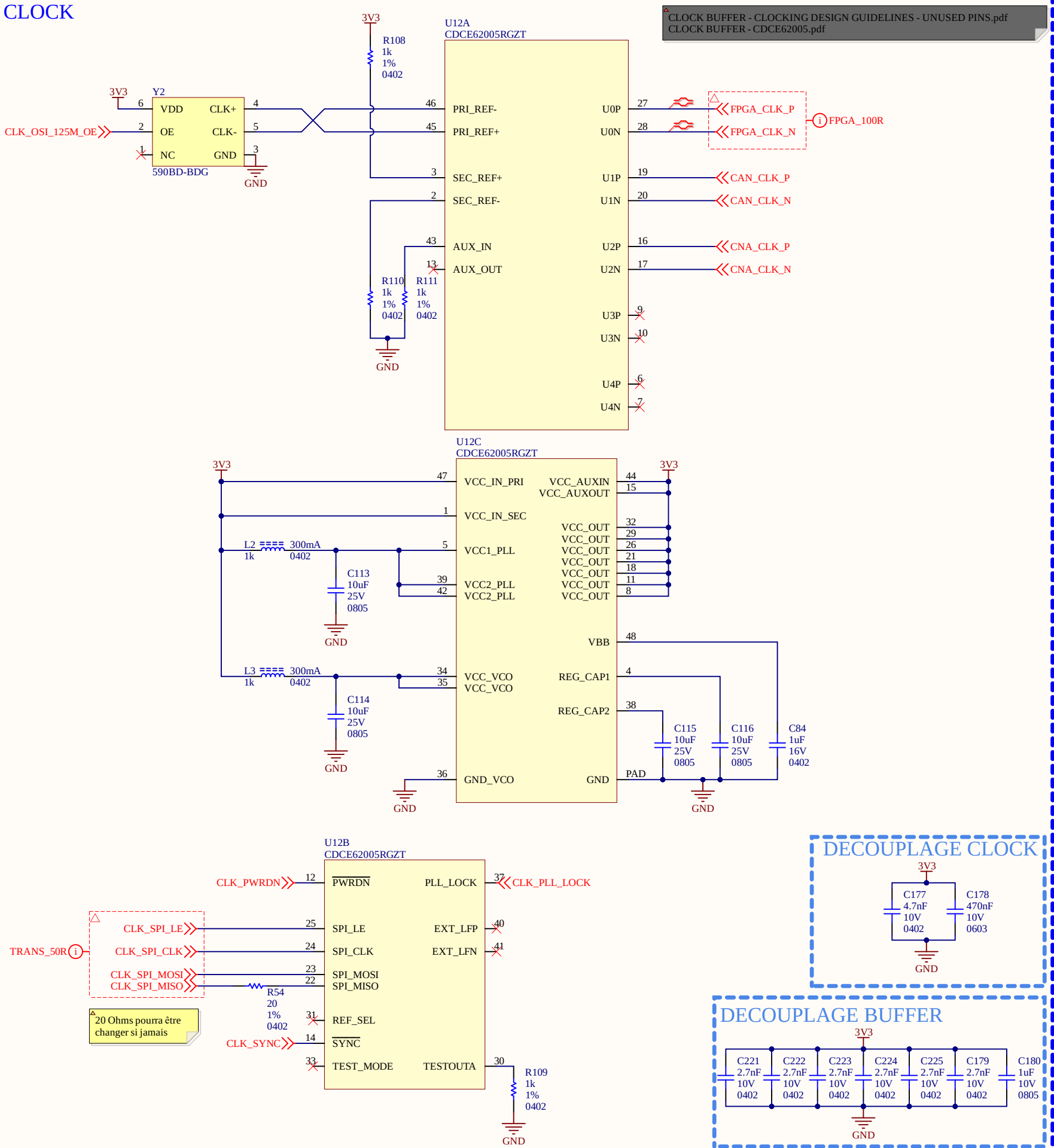


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Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	1 of 17
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_TITLE.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



Sheet Name		ALIM	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	3 of 17
Filename		Designers	
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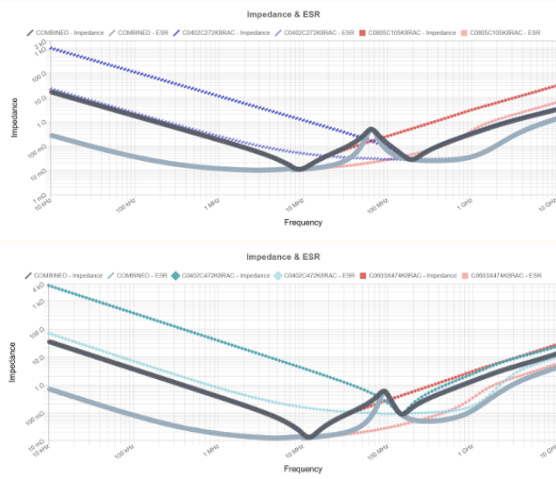
CLOCK



CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf
CLOCK BUFFER - CDCE62005.pdf

DECOUPLAGE CLOCK

DECOUPLAGE BUFFER

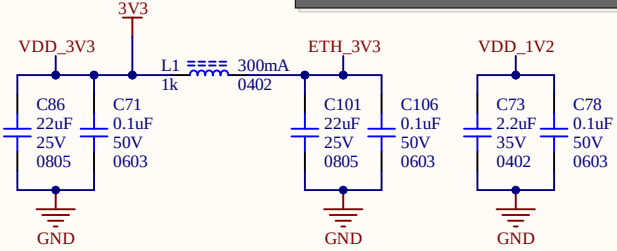


-Quand tu es en LVDS faut que branche le SEC à une résistance (CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf)
-CAN et CNA pas fait pour la pair différentiel puisqu'elle est directement dans le CAN et CNA

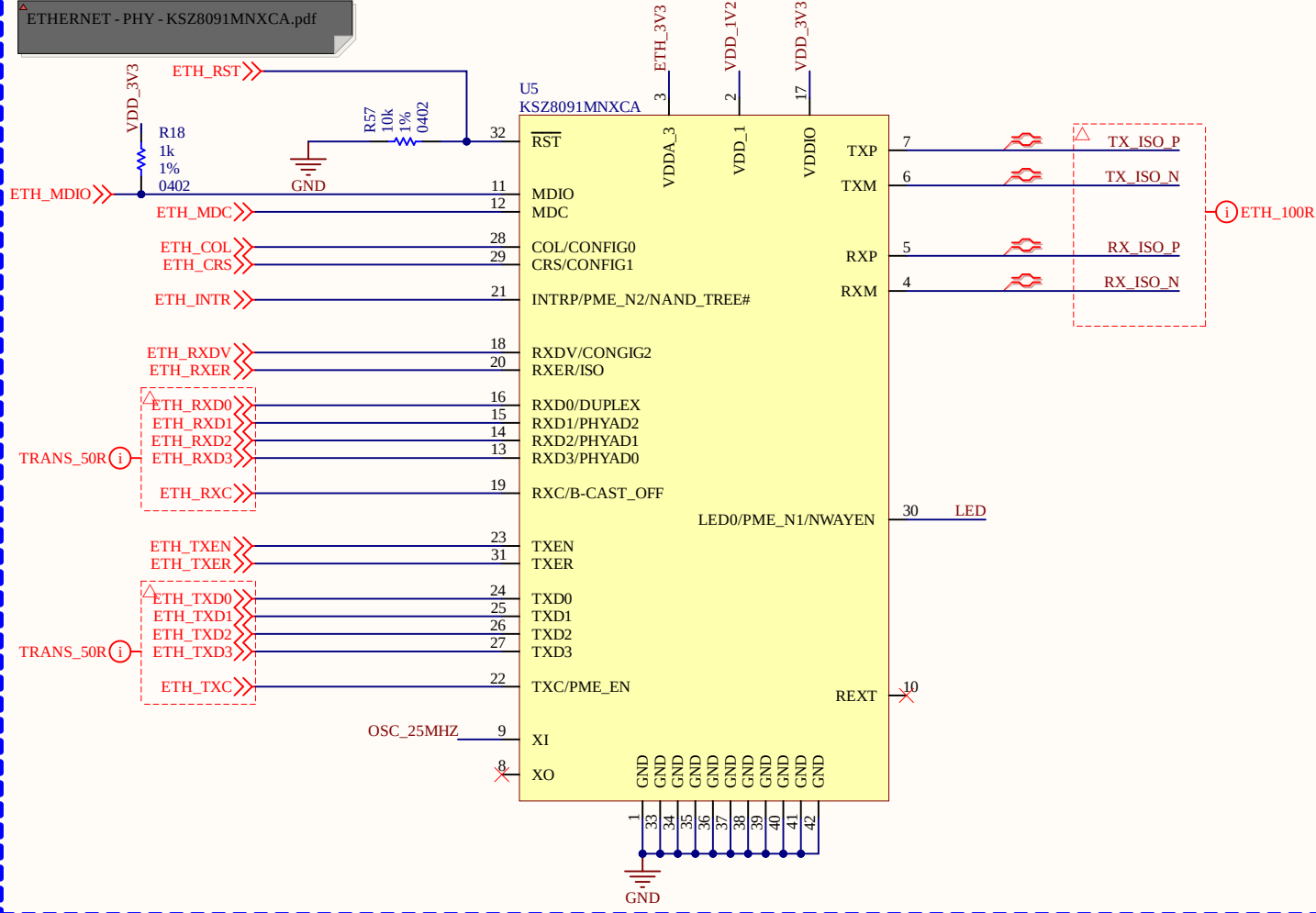
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Global Project			Pleiades		
Size	Group			Revision	
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Date	13/01/2026		Sheet	4	of 17
Filename				Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

ETHERNET
ALIMENTATION

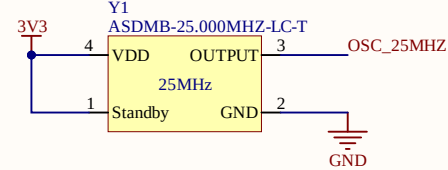
ETHERNET - PHY - KSZ8091MNXCA.pdf
P.37



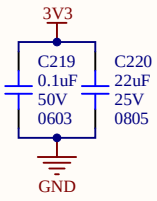
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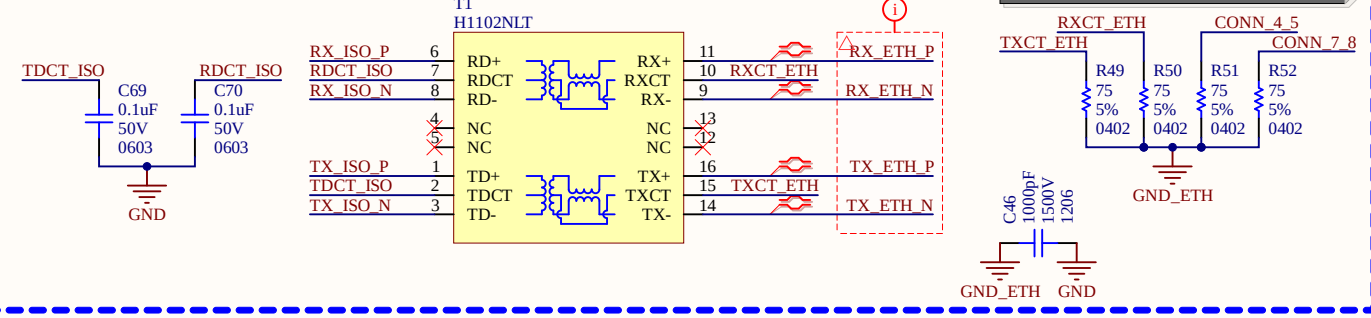
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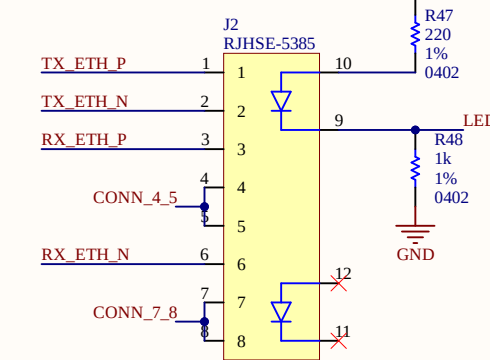
DECOUPLAGE



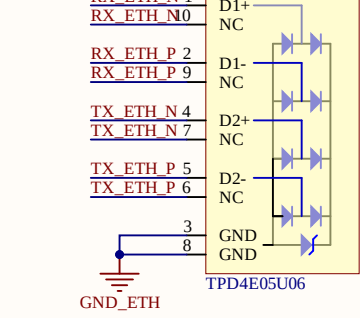
ISO



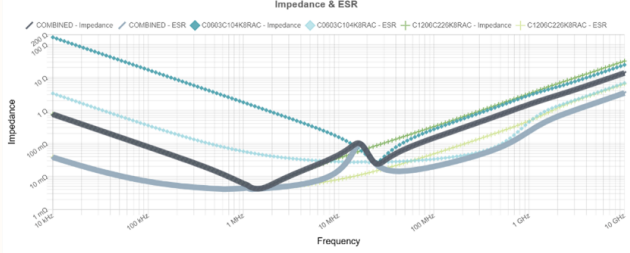
CONNECTEUR



ESD



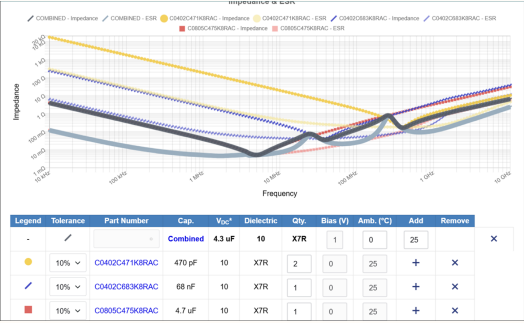
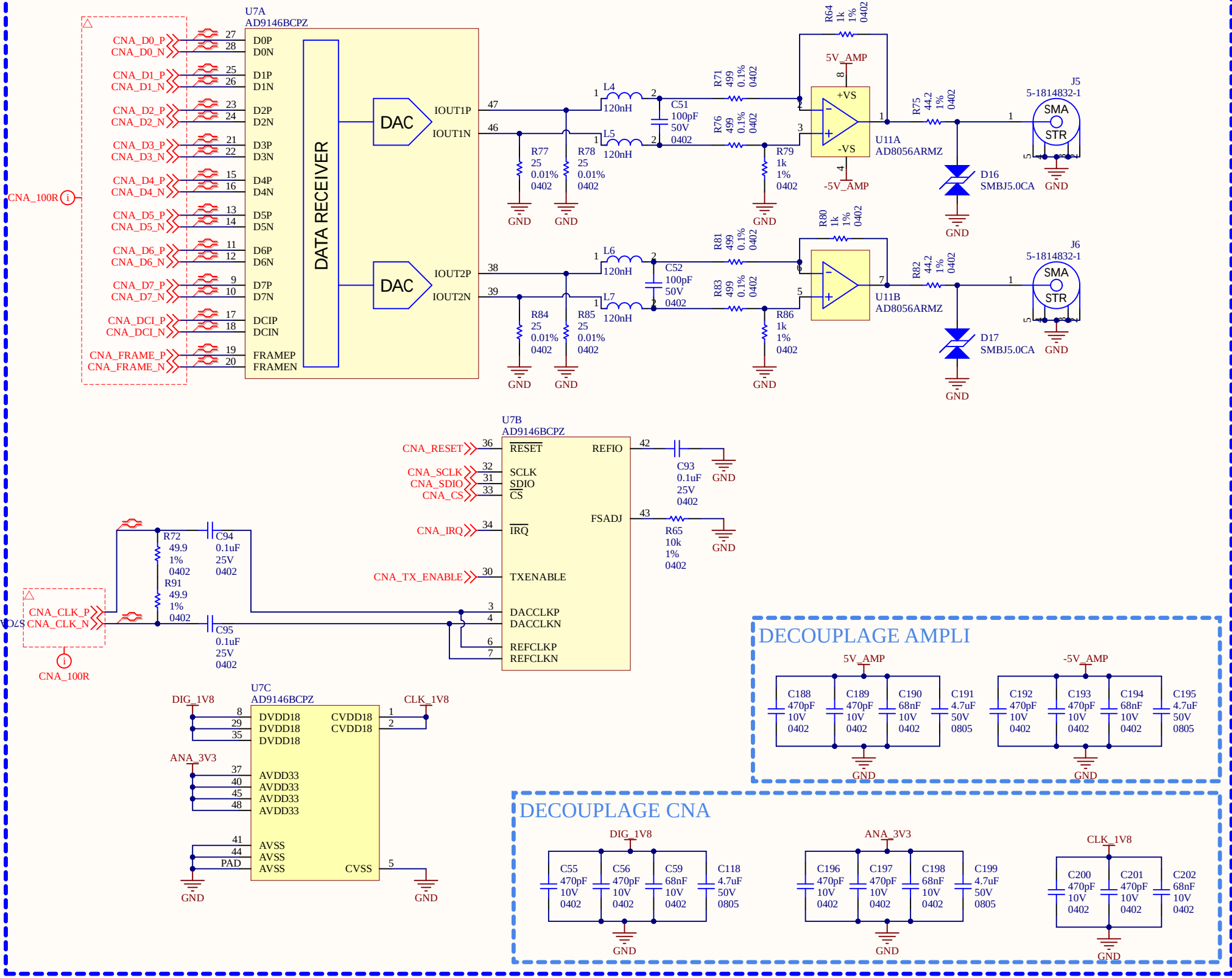
TIA 568A			
Pin #	Wire Color Legend		Signal
1	White/Green	TX+	
2	Green	TX-	
3	White/Orange	RX+	
4	Blue	TRD2+	
5	White/Blue	TRD2-	
6	Orange	RX-	
7	White/Brown	TRD3+	
8	Brown	TRD3-	



-Isolation du ETHERNET 1.5kV
-Ajout d'un condensateur de 1.5kV
-Ligne de transmission de 50R
-TX et RX voir le référence design pourquoi c'est ligne de transmission sans différentiel

Sheet Name			
ETHERNET			
Project Title			
S7CAS-PLEIADES-CARTE-MERE			
Global Project			
Pleiades			
Size	Group	Revision	
11x17		01	
Date	13/01/2026		Sheet 5 of 17
Filename	S7CAS-PLEIADES-CARTE-MERE_P01_ETHERNET.SchDoc		Designers Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard

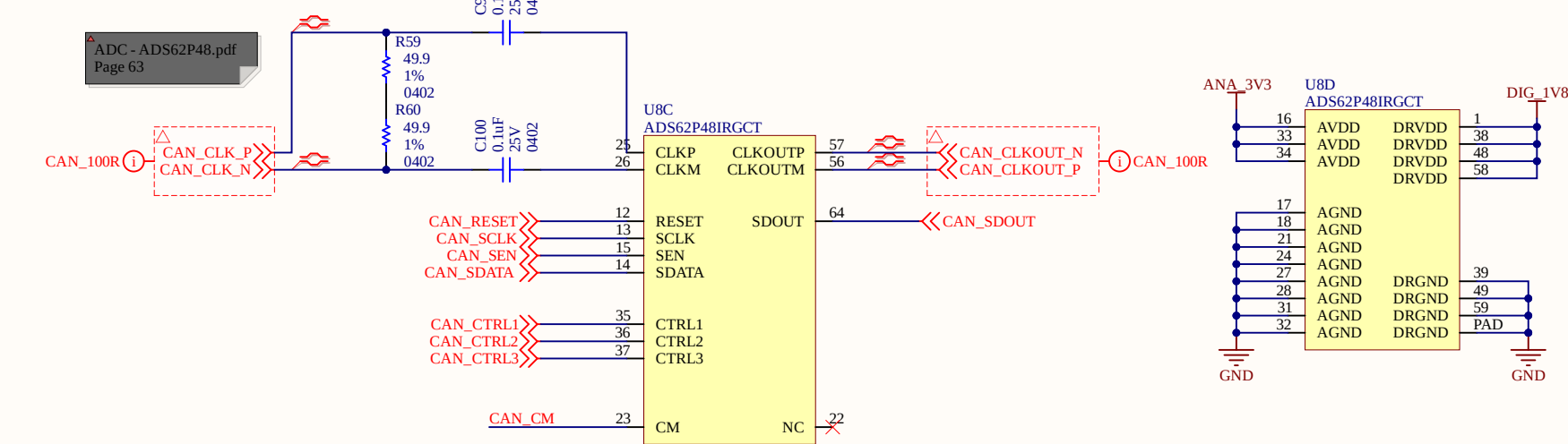
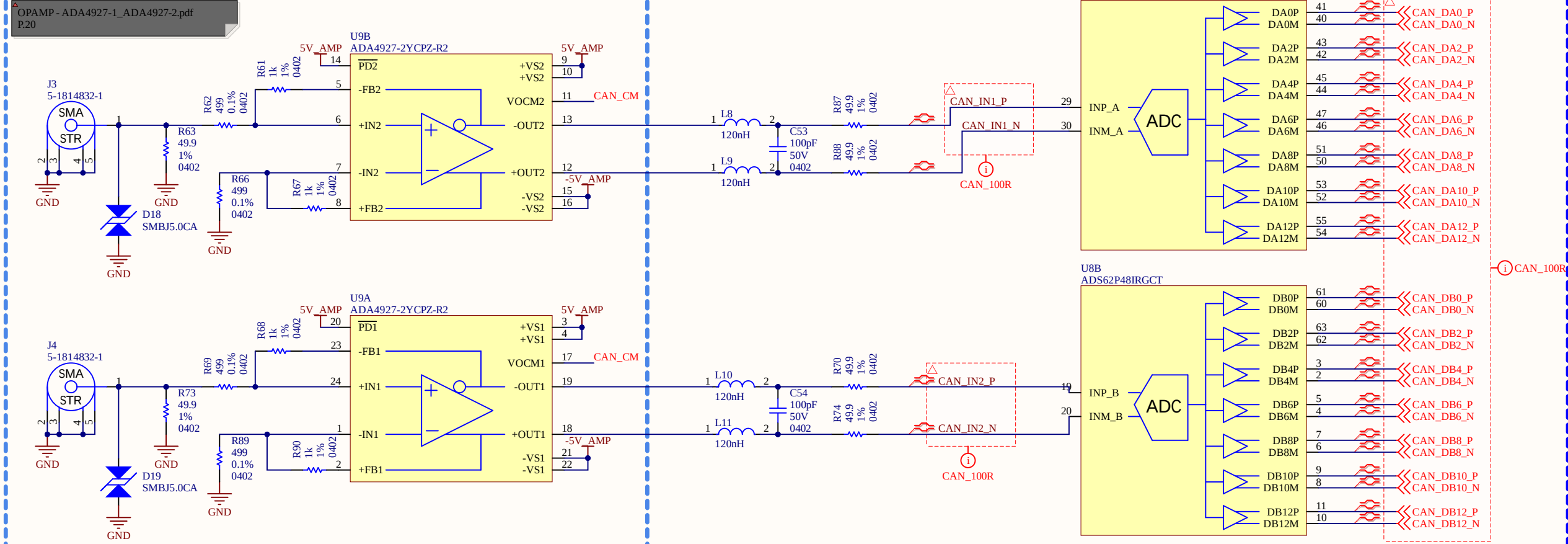
CNA (DAC)



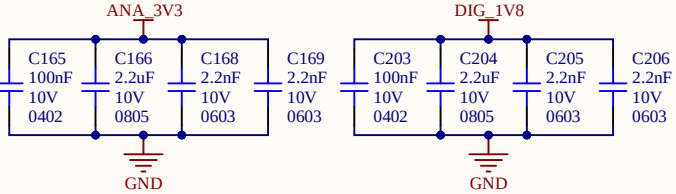
-200mbit/s en DDR LVDS
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Ampli-op aimerait un gain de 2 avec 1V à la sortie, il faut donc 0.5V en entree
-DAC sortie au maximum c'est 20mA, donc avec le 25ohms c'est 0.5V
-On réussi à avoir 1V à la sortie
-5ohms à la sortie à 40MHz donc 45ohms pour adapter
-Filtre passe bas de deuxième ordre (0 à 40MHz), point de coupure 45MHz = L et C
-Receiver Differential impedance 80 to 120ohms

Sheet Name		CNA	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
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Date	13/01/2026	Sheet	6 of 17
Filename		Designers	
S7CAS-PLEIADES-CARTE-MERE_P01_CNA.SchDoc		Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	

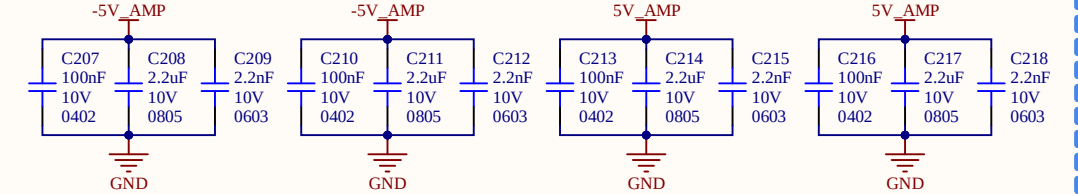
CAN (ADC)
DECOUPLAGE ANALOG



DECOUPLAGE CAN



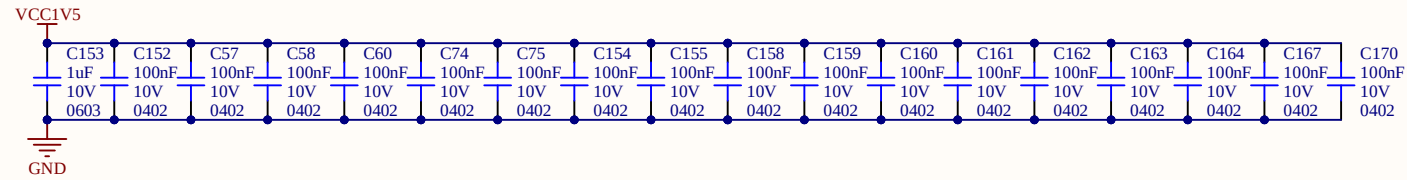
DECOUPLAGE AMPLI



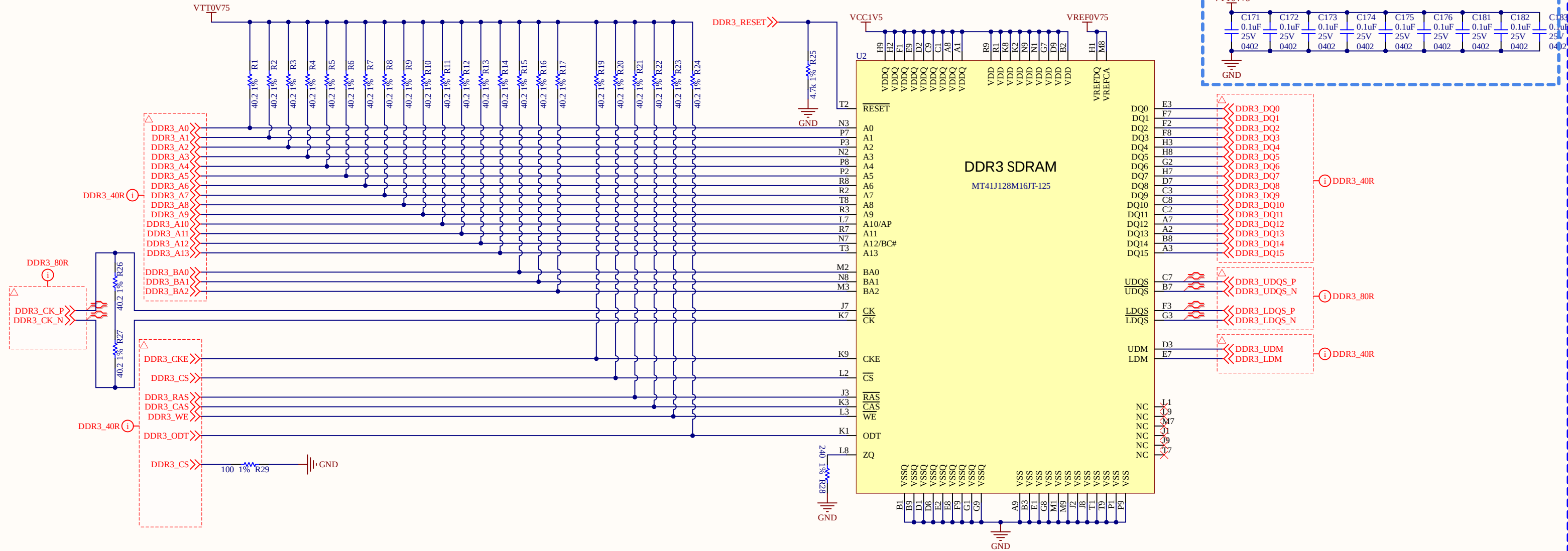
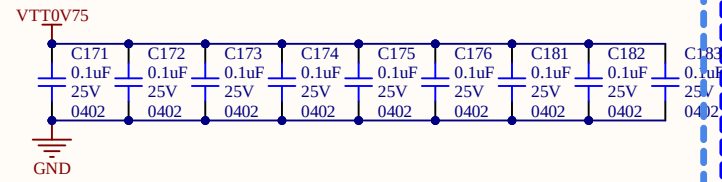
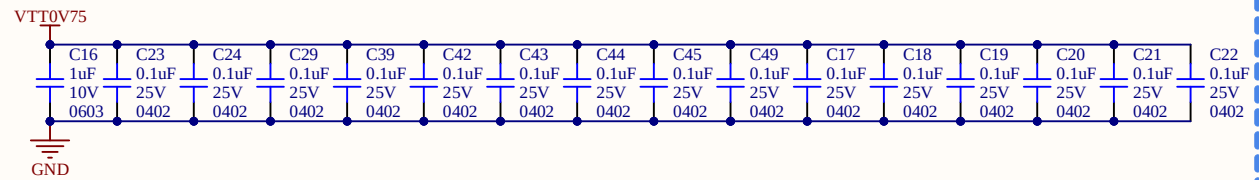
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Global Project			Pleiades		
Size	Group			Revision	
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Date	13/01/2026		Sheet	7	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_CAN.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

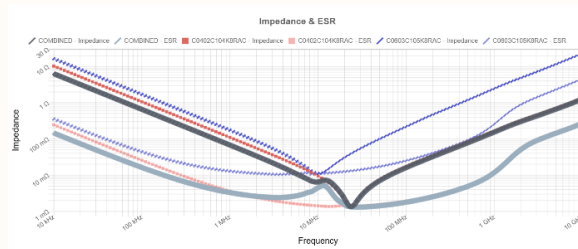
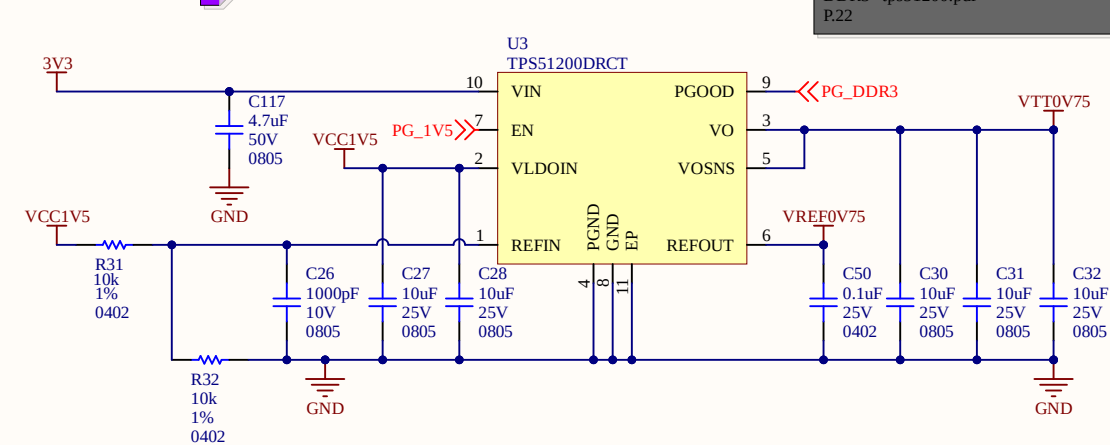
DECOUPLAGE



DECOUPLAGE



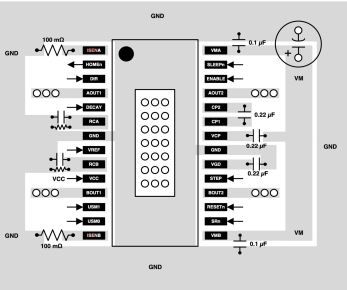
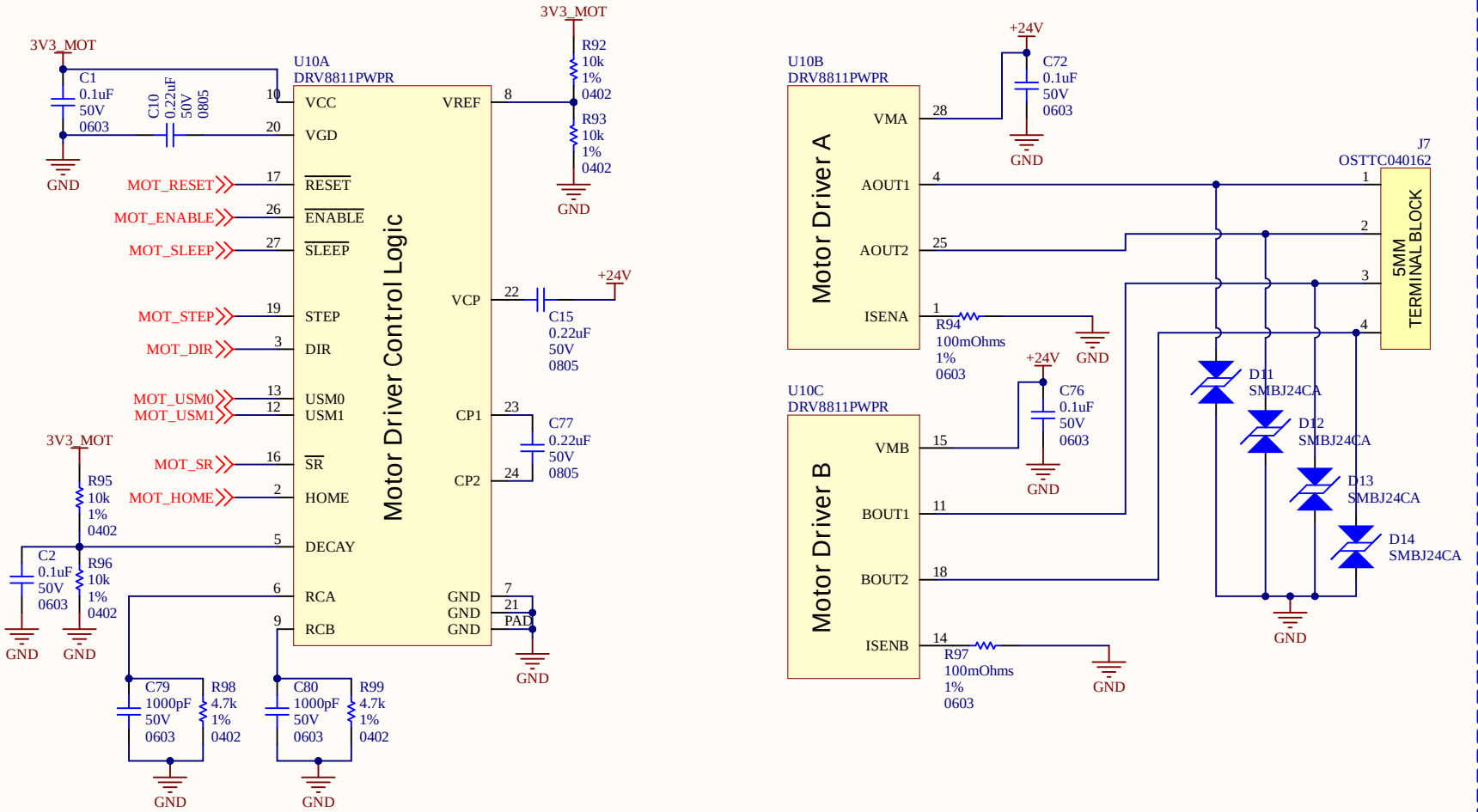
ALIM DDR3



- Ligne de transmission de 40R
- Resistance parallèle sur la ligne de transmission de 40.2Ohms
- Clock en pair différentiel 80R
- Découplage il y a 18 pins donc on peut en mettre 18 condensateurs
- Dans un cours on s'est fait dire que c'est un condensateur par résistance de VTT

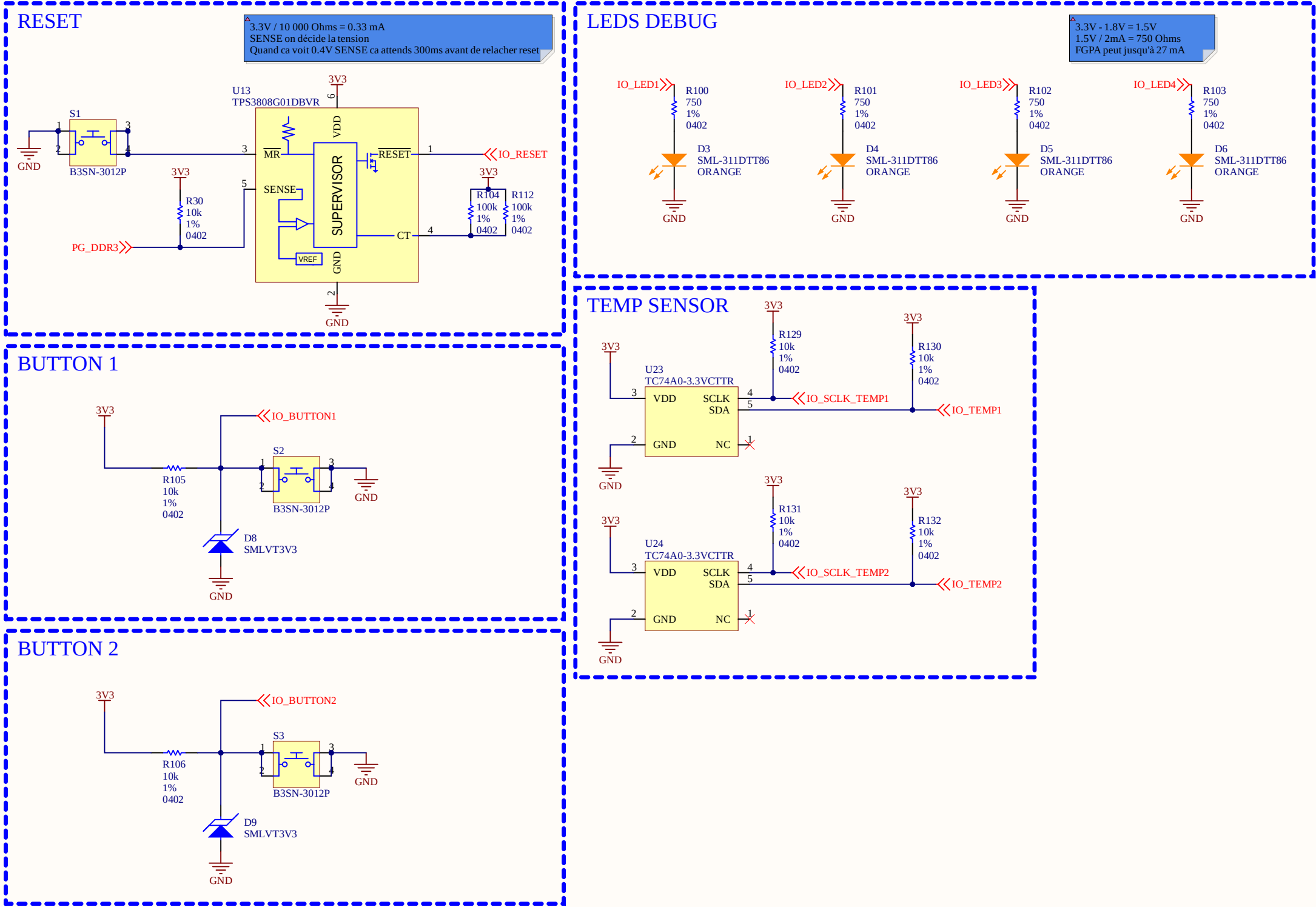
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Global Project			Pleiades		
Size	Group			Revision	
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Date	13/01/2026		Sheet	8	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_DDR3.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

MOTOR



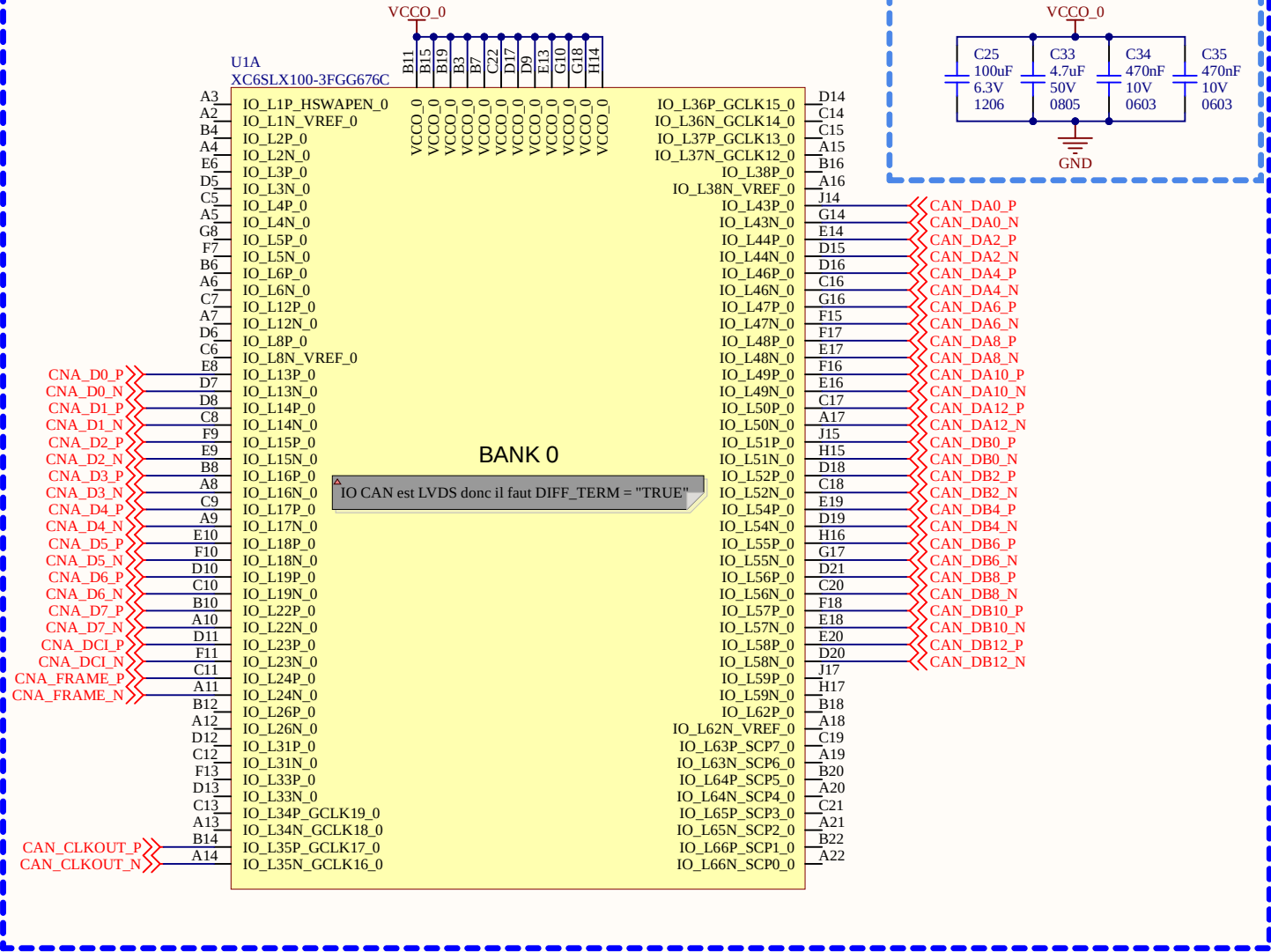
-Tous sur la BANK1
-ISENS = 1.65/(8x1,9A) = 0,108 Ohms

Sheet Name			MOTORS		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
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11x17	01			1.00	
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Filename				Designers	
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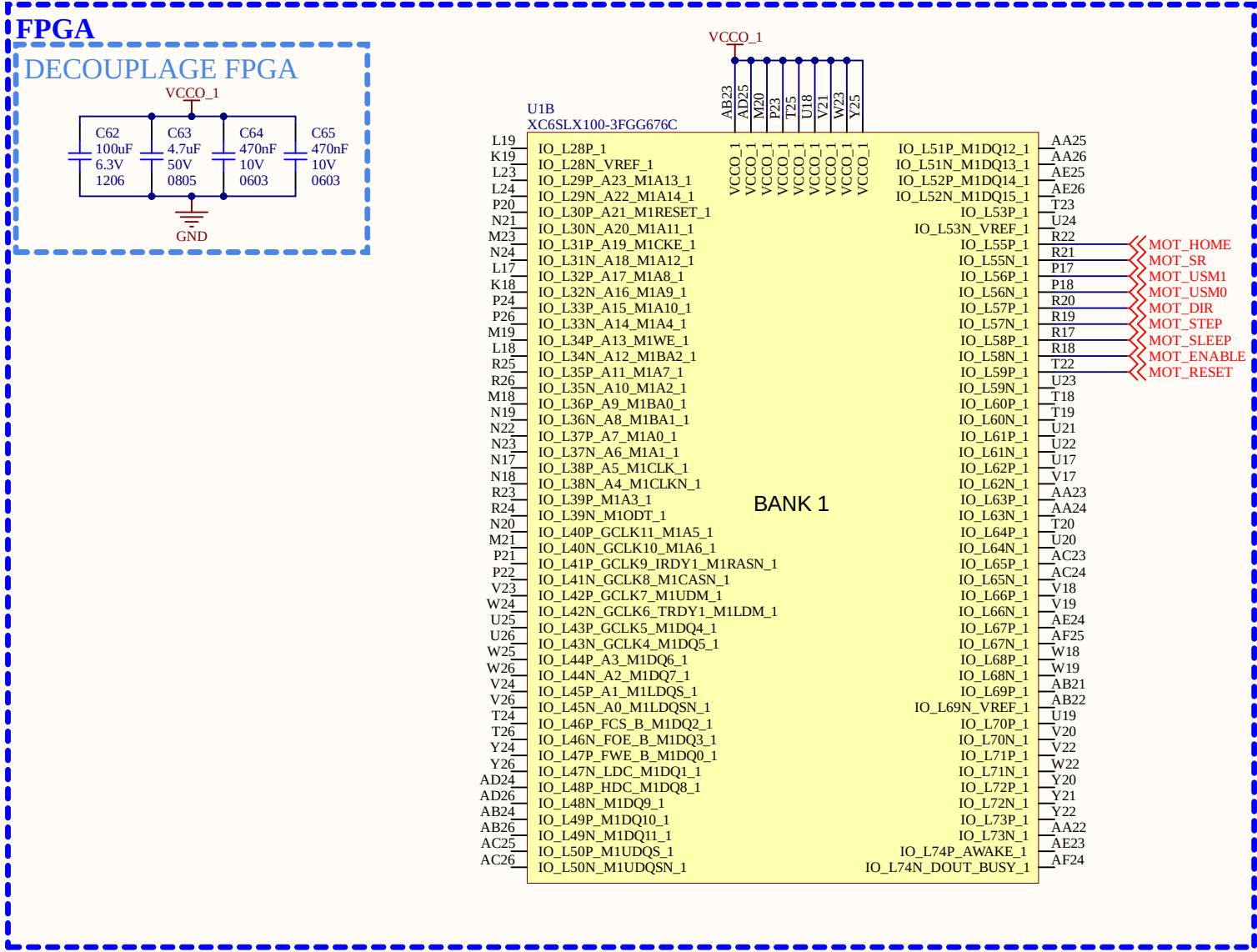
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Global Project			Pleiades		
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Date	13/01/2026		Sheet	10	of 17
Filename			Designers		
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FPGA

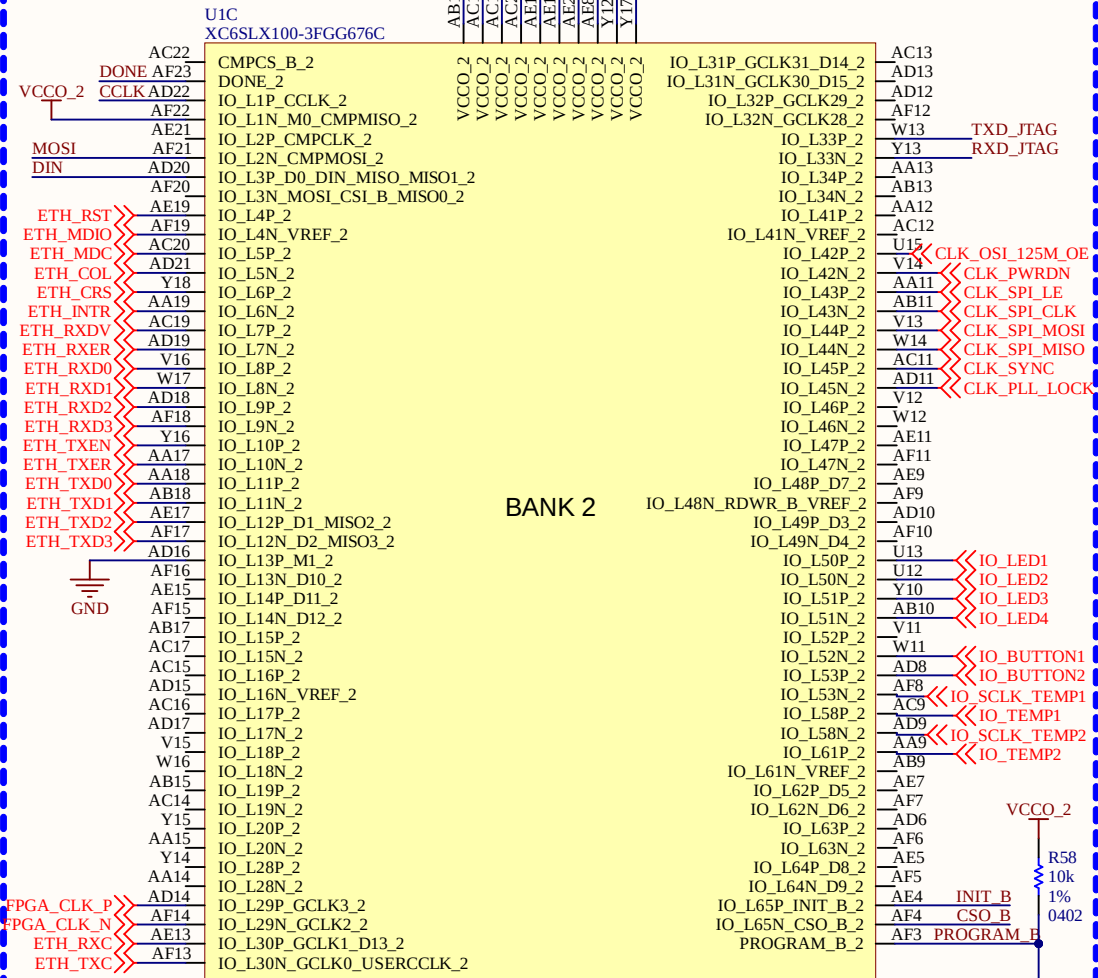


-LVDS seulement sur BANK0 et BANK2
-Toute les CAN et CNA sont mit sur cette BANK

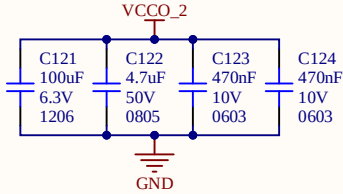
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Global Project			Pleiades		
Size	Group			Revision	
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Date	13/01/2026		Sheet	11	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		



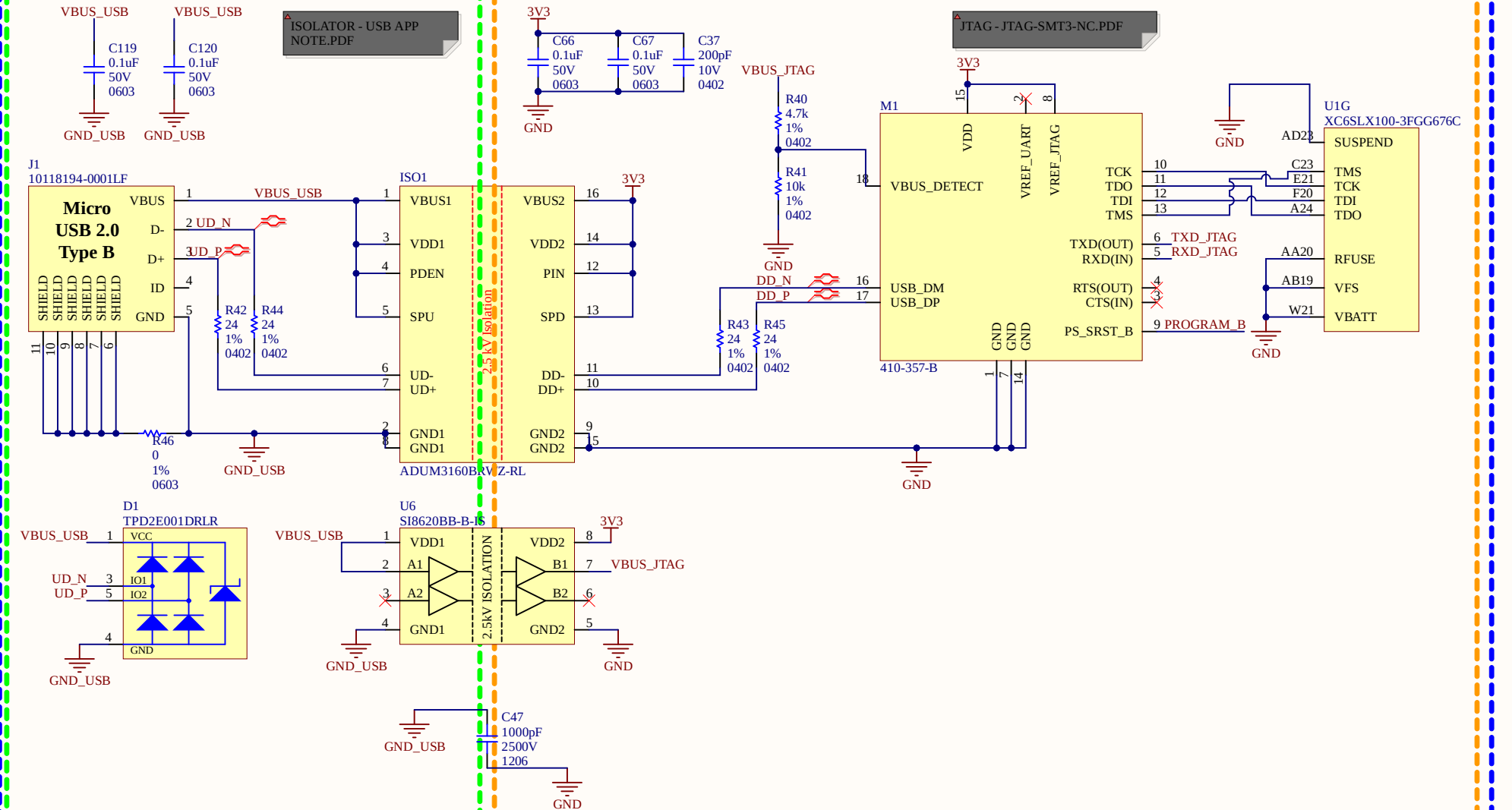
FPGA



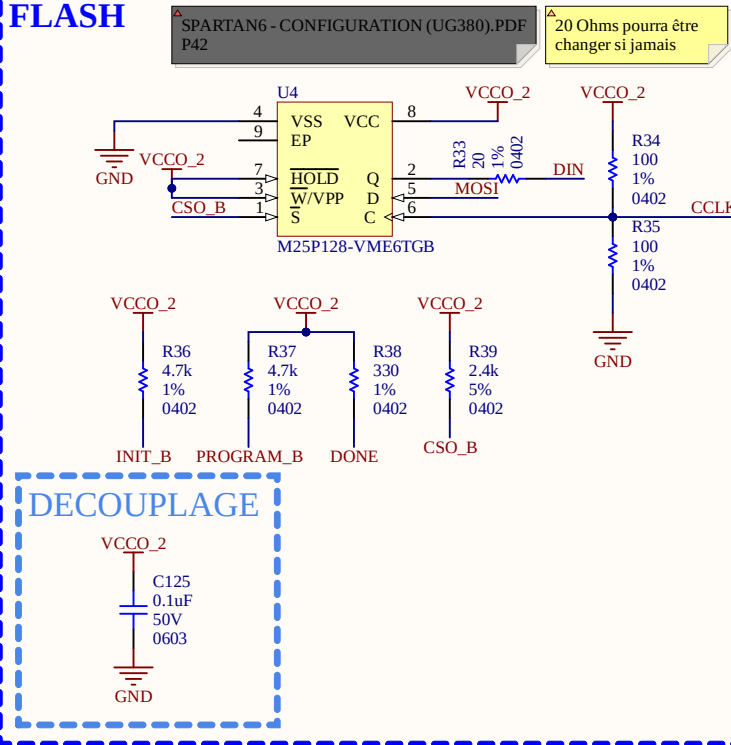
DECOUPLAGE FPGA



USB



FLASH

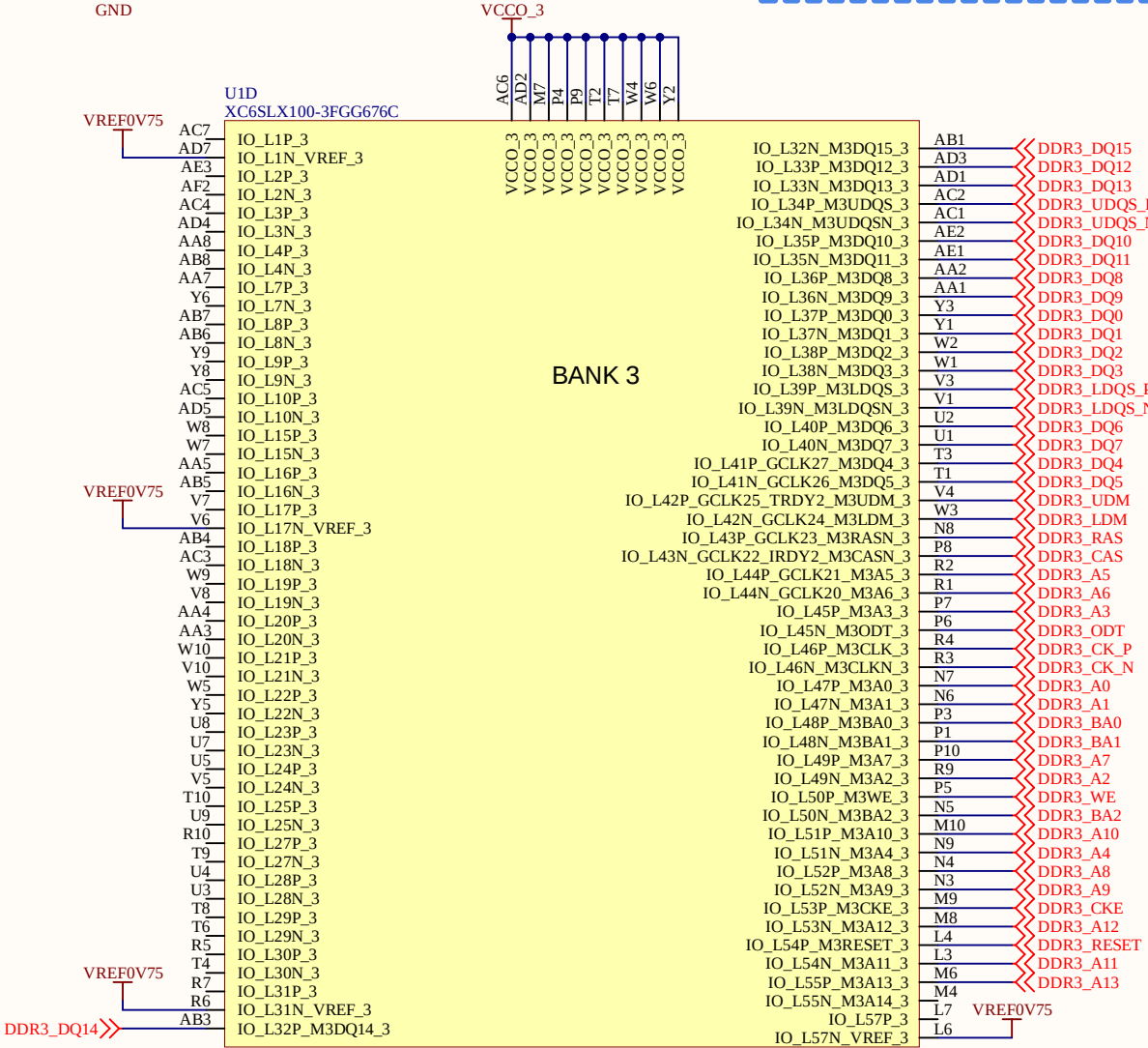
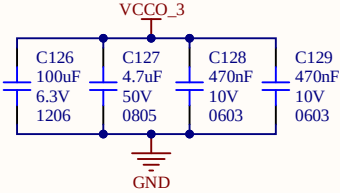
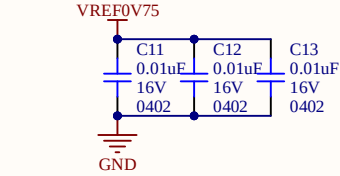


- Ajout de ESD pour la protection
 - Protection isolé pour le VBUS
 - Condensateur 2.5kV pour isolé les GND (même force que les isolateurs)
 - Paires différentiel sur le D+ et D-, avec des résistance de 24ohms (comme l'exemple)
 - Isolation du côté USB au lieu d'être du côté board (plus petit comme chip à partager du ISO de 6 fils)
 - If AES encryption is unused, VBATT can be tied to either, VCCAUX or ground, or left unconnected.
 - VFS and RFUSE are tied to ground because we are not using Efuses
- M0 et M1 choisi pour avoir le mode SPI
 - Ajout d'une résistance de 20 ohms sur Q
 - Toute la logique de code est sur la BANK ainsi que toute la programmation

Sheet Name			FPGA - BANK 2		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date		Sheet		of	
13/01/2026		13		17	
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

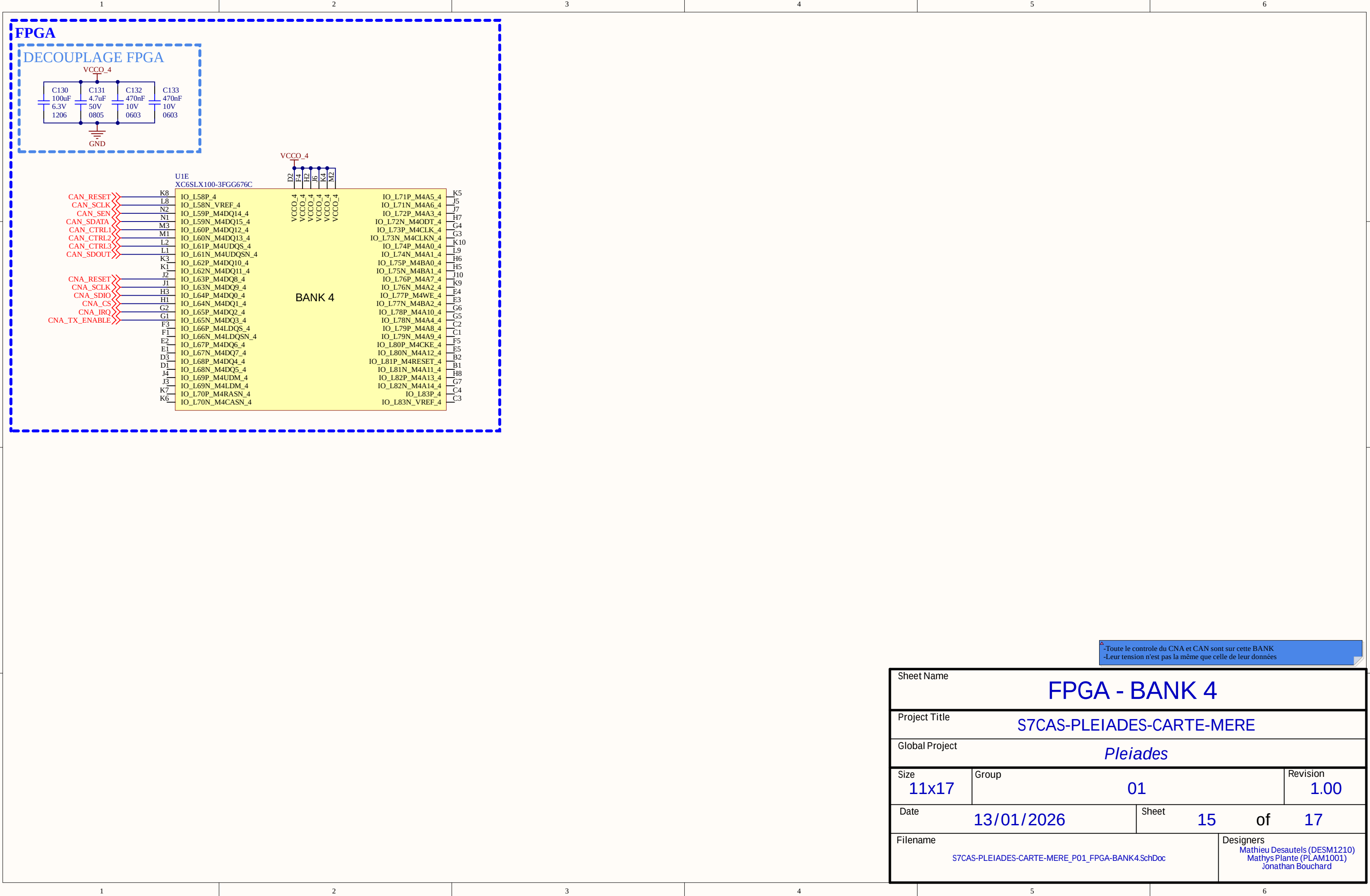
FPGA

DECOUPLAGE FPGA



-Toute la DDR3 est sur cette BANK

Sheet Name			FPGA - BANK 3		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
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Date	13/01/2026		Sheet	14	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK3.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		



-Toute le controle du CNA et CAN sont sur cette BANK
-Leur tension n'est pas la même que celle de leur données

Sheet Name			FPGA - BANK 4		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	15	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK4.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

U1F
XC6SLX100-3FPG676C

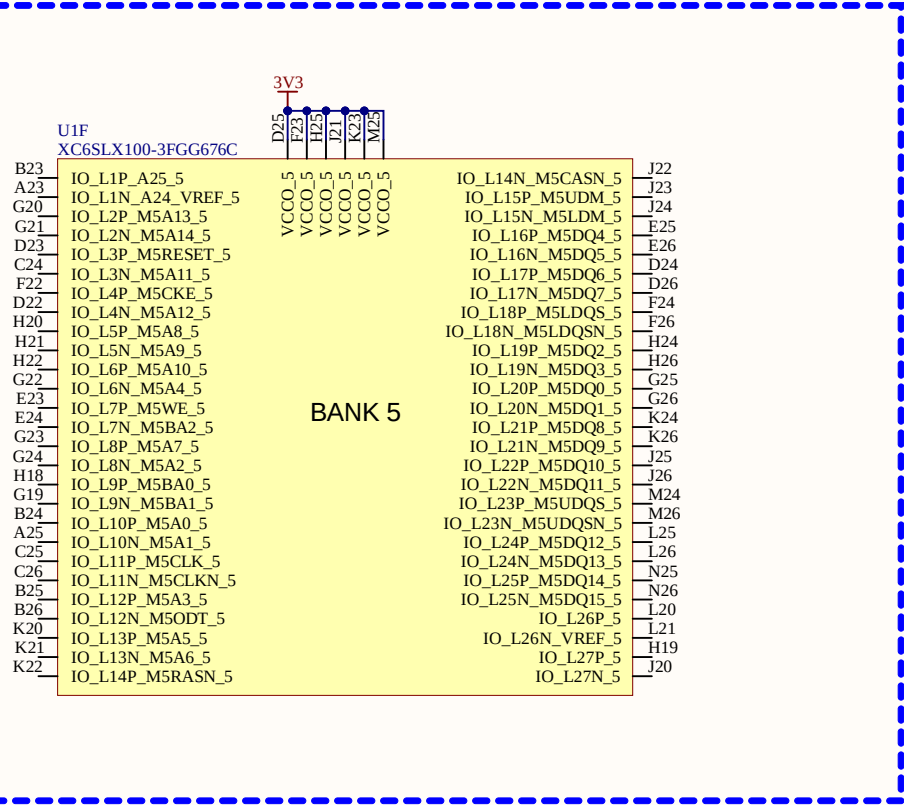
3V3

D25 F23 H23 J21 K23 M25

VCCO_5 VCCO_5 VCCO_5 VCCO_5 VCCO_5 VCCO_5

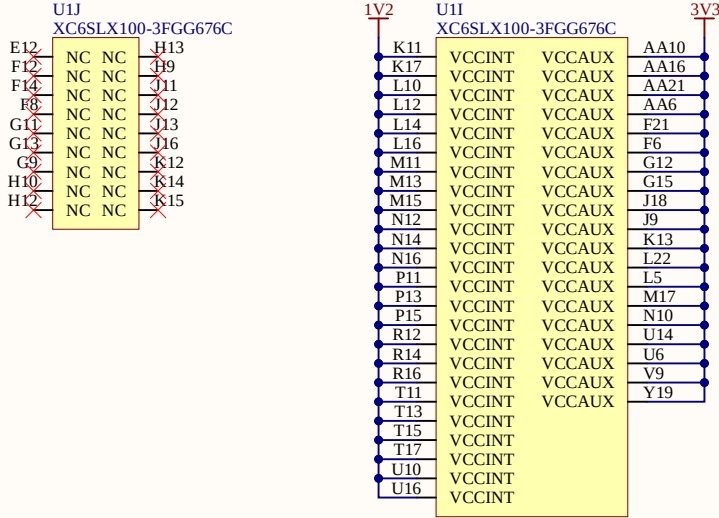
BANK 5

B23	IO_L1P_A25_5	IO_L14N_M5CASN_5	J22
A23	IO_L1N_A24_VREF_5	IO_L15P_M5UDM_5	J23
G20	IO_L2P_M5A13_5	IO_L15N_M5LDM_5	J24
G21	IO_L2N_M5A14_5	IO_L16P_M5DQ4_5	E25
D23	IO_L3P_M5RESET_5	IO_L16N_M5DQ5_5	E26
C24	IO_L3N_M5A11_5	IO_L17P_M5DQ6_5	D24
F22	IO_L4P_M5CKE_5	IO_L17N_M5DQ7_5	D26
D22	IO_L4N_M5A12_5	IO_L18P_M5LDQS_5	F24
H20	IO_L5P_M5A8_5	IO_L18N_M5LDQSN_5	F26
H21	IO_L5N_M5A9_5	IO_L19P_M5DQ2_5	H24
H22	IO_L6P_M5A10_5	IO_L19N_M5DQ3_5	H26
G22	IO_L6N_M5A4_5	IO_L20P_M5DQ0_5	G25
E23	IO_L7P_M5WE_5	IO_L20N_M5DQ1_5	G26
E24	IO_L7N_M5BA2_5	IO_L21P_M5DQ8_5	K24
G23	IO_L8P_M5A7_5	IO_L21N_M5DQ9_5	K26
G24	IO_L8N_M5A2_5	IO_L22P_M5DQ10_5	J25
H18	IO_L9P_M5BA0_5	IO_L22N_M5DQ11_5	J26
G19	IO_L9N_M5BA1_5	IO_L23P_M5UDQS_5	M24
B24	IO_L10P_M5A0_5	IO_L23N_M5UDQSN_5	M26
A25	IO_L10N_M5A1_5	IO_L24P_M5DQ12_5	L25
C25	IO_L11P_M5CLK_5	IO_L24N_M5DQ13_5	L26
C26	IO_L11N_M5CLKN_5	IO_L25P_M5DQ14_5	N25
B25	IO_L12P_M5A3_5	IO_L25N_M5DQ15_5	N26
B26	IO_L12N_M5ODT_5	IO_L26P_5	L20
K20	IO_L13P_M5A5_5	IO_L26N_VREF_5	L21
K21	IO_L13N_M5A6_5	IO_L27P_5	H19
K22	IO_L14P_M5RASN_5	IO_L27N_5	J20

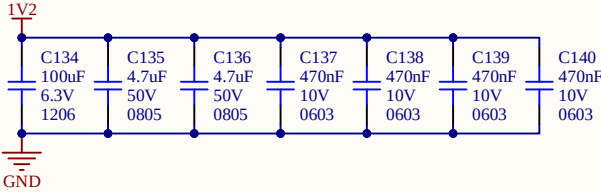


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Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
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Date	Sheet		of		
13/01/2026	16		17		
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK5.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA



DECOUPLAGE VCCINT



DECOUPLAGE VCCAUX

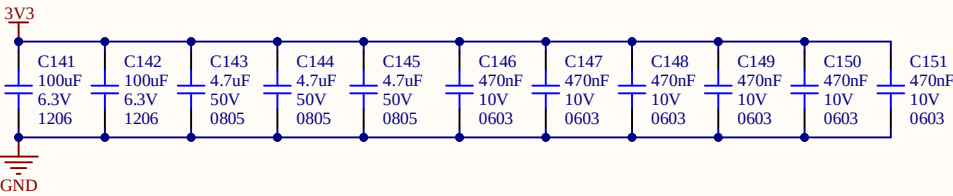


Table 2-1: Required PCB Capacitor Quantities per Device ⁽¹⁾⁽³⁾ (Continued)																	
Package	Device (XCES)	V _{CCINT} In µF	V _{CCAUX} In µF	V _{CC0} Bank 0 In µF	V _{CC0} Bank 1 In µF	V _{CC0} Bank 2 In µF	V _{CC0} Bank 3 In µF	V _{CC0} Bank 4 In µF	V _{CC0} Bank 5 In µF	Total (2)							
		100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	
FG(G)484	LX75	1	2	3	1	2	4	1	1	2	1	1	3	1	1	3	33
FG(G)484	LX75T	1	2	3	1	2	4	1	1	2	1	1	3	1	1	3	32
FG(G)484	LX100	1	2	4	1	2	4	1	1	2	1	1	3	1	1	3	33
FG(G)484	LX100T	1	2	4	1	2	4	1	1	2	1	1	3	1	1	3	33
FG(G)484	LX150	2	3	6	1	2	4	1	1	2	1	1	3	1	1	3	37
FG(G)484	LX150T	2	3	6	1	2	4	1	1	2	1	1	3	1	1	3	37
FG(G)676	LX45	1	1	2	1	2	5	1	1	3	1	1	3	1	1	4	33
FG(G)676	LX75	1	2	3	2	3	6	1	1	3	1	1	3	1	1	2	45
FG(G)676	LX75T	1	2	3	1	2	5	1	1	3	1	1	3	1	1	2	40
FG(G)676	LX100	1	2	4	2	3	6	1	1	3	1	1	3	1	1	2	46
FG(G)676	LX100T	1	2	4	1	2	5	1	1	3	1	1	3	1	1	2	41
FG(G)676	LX150	2	3	6	2	3	6	1	1	3	1	1	3	1	1	2	50
FG(G)676	LX150T	2	3	6	1	2	5	1	1	3	1	1	2	1	1	2	45
FG(G)900	LX100T	1	2	4	2	3	6	1	1	3	1	1	3	1	1	2	47
FG(G)900	LX150	2	3	6	2	4	7	1	1	5	1	1	5	1	1	2	57
FG(G)900	LX150T	2	3	6	2	3	7	1	1	4	1	1	4	1	1	2	54

Sheet Name

FPGA - ALIM

Project Title

S7CAS-PLEIADES-CARTE-MERE

Global Project

Pleiades

Size

11x17

Group

01

Revision

1.00

Date

13/01/2026

Sheet

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Filename

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